Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (IT) Examination

December 2016

IT302.02/IT302.01/IT302 Design & Analysis of Algorithm

Date: 19.12.2016, Monday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Attempt the following questions. **[07]**

1. How many times is the comparison *i* <= *n* performed in the following program? [01]

```
int i = 10, n = 100;
main(){
        while (i <= n)
         {
              i = i+1;
              n = n-1;
         }
}
```

2. What is the worst case of an algorithm? [01]
3. Which of the algorithm design technique is used in the quick sort algorithm? [01]
4. Find out total number of primitive operations for following code. [01]

```
int Array_Min(A, N)
{      current_Min ← A[0]
       for i ← 1 to n − 1 do
        {
            if current_Min > A[i] then
                   current_Min ← A[i]
           i ← i+1
        }
return current_Min
}
```

5. What is the time complexity of Strassen's matrix multiplication algorithm? [01]
6. Explain general structure of divide and conquer approach. [02]

Q-2 (a) Write general recurrence equation for divide and conquer approach. Explain and analyze best case, average case and worst case time complexity of any one algorithm using divide and conquer approach. **[06]**

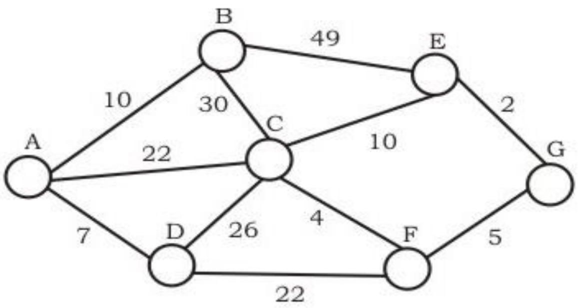
Q-2 (b) Student has to complete 12 courses for their degree. There are six compulsory courses: Basic and Advanced Mathematics, Basic and Advanced Physics and Basic and Advanced Electronics. Student also has to complete six Optional Courses. Each course takes one semester to complete. There are some constraints because of prerequisites. **[04]**

- For Mathematics, Physics and Electronics, the Basic course must be completed before starting the advanced course.
- Advanced Physics must be completed before starting Basic Electronics.
- Advanced Mathematics must be completed before starting Advanced Physics.
- The Optional Courses can be done in any order, but no more than two Optional Courses can be taken in a semester.

Given these constraints, answers the following questions.

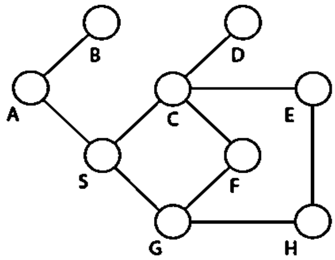
1) Decide and design suitable algorithmic approach to solve the above problem.
2) What is the minimum number of semesters that Student needs to complete 12 courses?

Q-2 (c) Construct minimum spanning tree using Prim's algorithm. Consider node A as a starting node. Note down the sequence of edges represents order in which the edges would be added to construct the minimum spanning tree. **[04]**

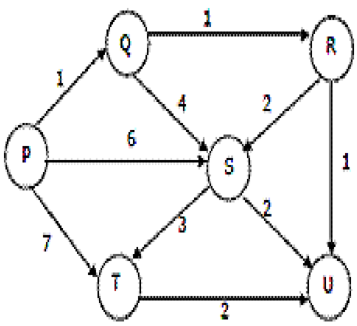


OR

Q-2 (b) Define Graph. List out different techniques for graph traversal and apply them on given graph with source vertex S. **[04]**



Q-2 (c) Apply Dijkstra's single source shortest path algorithm on the following edge weighted directed graph with vertex P as the source vertex. Also note down the sequence of the vertex visited. **[04]**



Q-3 (a) Explain characteristics of greedy algorithm. **[04]**

(b) Attempt the following questions. (Any Two) **[10]**

1. Solve the following Knapsack Problem using Greedy Algorithm. There are five items whose weights and prices are given in following table. The capacity of the knapsack is 100 kg.

Item	I1	I2	I3	I4	I5
Weight (kg)	40	30	10	50	20
Price (Rs.)	40	66	20	60	30

2. Define N queen problem. Draw pruned state space tree for 4-queen problem.

3. Use branch and bound to solve the assignment problem (Minimization) with the following cost matrices.

	J1	J2	J3	J4
P1	11	12	18	40
P2	14	15	13	22
P3	11	17	19	23
P4	17	14	20	28

SECTION-II

Q-4 Attempt the following questions. **[07]**

1. If $T(n) = n\sqrt{n}$ then [01]

(a) $T(n) = O(n)$ (b) $T(n) = O(n \log n)$

(c) $T(n) = O(n^2)$ (d) None of the above

2. Let W (n) and A (n) denote respectively, the worst case and average case running time of an algorithm executed on an input of size n. which of the following is ALWAYS TRUE? [01]

(a) $A(n) = \Omega(W(n))$ (b) $A(n) = \theta(W(n))$

(c) $A(n) = O(W(n))$ (d) None of above

3. Let $f(n) = n(1+\sin(n))$ and $g(n) = n$, where n is a positive integer. Which of the following statements is/are correct? [01]

I. $f(n)=O(g(n))$ II. $f(n)=\Omega(g(n))$

(a) Only I (b) Only II

(c) Both I and II (d) Neither I nor II

4. Construct the table for calculating Binomial Coefficient for Compute (n, k) = (7,5) [02]

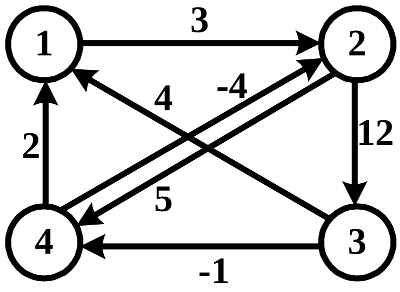
5. Arrange the given terms in increasing order of their time complexity. [02]

$O(\sqrt{n})$, $O(n^{(3/2)})$, $O(lg(lgn))$, $O(n)$, $O(n^2 lgn)$, $O(2^n)$, $O(n!)$, $O(nlgn)$

Q-5 (a) What is Asymptotic Notation? Discuss importance of asymptotic notations in design and analysis of algorithm. Also explain different asymptotic notations with neat sketch. **[04]**

Q-5 (b) Discuss basic idea of Rabin-Karp string matching algorithm. What is the running time complexity of this algorithm? **[05]**

Q-5 (c) A directed weighted graph is given as below. Find all pair shortest path using Floyd Algorithm. **[05]**



OR

Q-5 (b) State Master Theorem and Solve following recurrence using Master Theorem. **[05]**

1. $T(n) = 2T(n/4) + n^{0.51}$
2. $T(n) = 4T(n/2) + n^2$

Q-5 (c) (1) Define and show relationship between following classes. **[05]**

P , NP, NP Hard, NP Complete

(2) What is reduction in complexity class?

Q-6 Attempt the following questions. (Any Two) **[14]**

1. Find Upper bound of following recurrence equation using Recurrence Tree Method.
   1. $T(n) = 4T(n/2) + n^2$
   2. $T(n) = T(n/10) + T(9n/10) + n$

2. Using algorithm find an optimal parenthesis of a matrix chain product whose sequence of dimension is (3,2,5,3,4,4)

3. Find any one Longest Common Subsequence of given two strings using Dynamic Programming. Two amino-acid sequences H E A G A W G H E E and P A W H E A E. The corresponding amino-acids are A: Alanine, E: Glutamic acid, G: Glycine, H: Histidine, P: Proline, and W: Tryptophan

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